

MINH (BEN) PHAM

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EDUCATION

Bachelor of Science in Computer Science,
University of California, Riverside—GPA: 3.6/4.0

June 2026

PROFESSIONAL SUMMARY

Analytics & ML: Python ETL pipelines on 540K+ transactions; RFM segmentation, cohort analysis, NLP classification.

Business Impact: Projected 1.85M–2.8M Euros revenue via retention analysis; +15% vulnerability categorization accuracy at Webacy.

Technical Depth: Python, SQL, Tableau, scikit-learn; XGBoost, SHAP, SMOTE for predictive modeling and fraud detection.

Background: Seeking Data Analyst / Data Scientist roles - June 2026 - open to full-time and internship opportunities and relocation.

EXPERIENCE

AI/ML Researcher | ACM@UCR - Data, AI, Statistic | Riverside, CA |

March 2026 – June 2026

- Developed a credit card fraud detection pipeline on a **284K-transaction dataset with a 0.17% fraud rate**, benchmarking KNN, Logistic Regression, and SMOTE-augmented models to analyze precision-recall tradeoffs under severe class imbalance.
- Trained a final XGBoost model with tuned detection sensitivity, achieving **91% precision** and **84% recall** on fraud cases.
- Addressed a severe **577:1 class imbalance** (fraud is rare) using **SMOTE** oversampling, boosting fraud detection rate from **63% to 92%**, and analyzing the tradeoff between catching more fraud vs. creating more false alerts.
- Used **SHAP** to explain which transaction features drove fraud predictions, making model decisions interpretable and surfacing the strongest fraud signal across the test set.

Data Analytics Extern | Webacy Web3 Security (via Extern) | Remote, CA |

July 2024 – August 2024

- Applied unsupervised clustering in Python to surface vulnerability patterns, improving categorization accuracy by **15%** and enabling consistent risk profiling across **1,000+** contracts.
- Isolated the **top 5 exploitable vulnerability classes**, redirecting remediation focus where breach risk was highest.
- Built heatmap visualizations to expose cross-vulnerability correlations that were invisible in raw tables, increasing risk-correlation visibility by **20%**, and presented security recommendations to the team and nontechnical stakeholders.
- Normalized and **cleaned** noisy datasets using statistical outlier detection, reducing outlier contamination by **12%** and improving clustering stability for downstream security audits across Webacy's contract dataset.

Course Grader & Teaching Staff (Software Construction) | UC Riverside | Riverside, CA |

September 2025 – June 2026

- Evaluated **50+** student projects per week on **CI/CD** pipelines, **unit testing**, and **Agile** workflows; provided targeted code reviews that measurably raise code quality and test coverage each quarter.
- Translated student performance data in **Excel** into plain-language progress reports for instructors, enabling early intervention for at-risk students and giving non-technical staff clear **visibility** into course health each quarter.
- Instructed **100+ students** each quarter on **Test-Driven Development (TDD)** and **Scrum**, leading weekly sprint reviews with VS Code and Git, which improved student adoption of TDD practices and earned positive feedback from most participants.

PROJECTS

Online Retail Customer Analytics Dashboard | [GitHub](#) | Python, Tableau |

March 2026

- Designed a Python **ETL pipeline** to standardize and clean **540K+** inconsistently formatted transactions across the retail dataset, resolving data quality issues that would have distorted downstream **RFM segmentation** and **cohort results**.
- Applied RFM segmentation in Python to the retail dataset, revealing that the top customer tier drove **65.2%** of revenue, directly informing which segments to prioritize in the retention strategy.
- Built cohort retention analysis with interactive **heatmap data visualization**, revealing **70–80%** churn after first purchase, signaling that first-purchase conversion was the single highest-leverage retention opportunity.
- Generated executive report with cohort and RFM recommendations projecting **1.85M–2.8M Euros** annual revenue impact through retention and expansion strategies.

GameDar—Gaming Forum With AI | [GitHub](#) | Python, Django, RESTAPI, SQL |

September 2025 – March 2026

- Integrated the IGDB API (Twitch) to seed a normalized **PostgreSQL** schema with real game metadata, then queried it using **complex joins** and **aggregations** to power ratings, sentiment, and vote aggregation across **9 aspect categories**.
- Built a review engine using **zero-shot text classification (BART)** to categorize feedback into 9 distinct aspects and aggregate ratings, sentiment, and top categories per game.
- Developed an AI sentiment pipeline via Hugging Face; categorized reviews with **65%+** confidence, giving users structured insight into exactly what aspects of a game reviewers praised or criticized.

TECHNICAL SKILLS

Languages & Libraries: Python (scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Seaborn, Matplotlib), R, SQL/PostgreSQL, C++, Hugging Face (DistilBERT, BERT, NLP Pipelines), MPI

Data & Analytics: Excel (Macros, Pivot Tables, VLOOKUP), Tableau, Cohort Analysis, RFM Segmentation, Sentiment Analysis, Data Visualization, Jupyter, Google Colab

Tools & Platforms: Supabase, Firebase (GCP), CI/CD, Bash, VS Code, Git, GitHub